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First edition, [YEAR]

[Dedication goes here.]

Act One — Clause 9.4.2 {#act-one-clause-942}

Day 0 — Earth, signing day

Chapter One — Clause 9.4.2

Forty-seven people in a room designed for thirty, and not one of them was happy about it.

Cole Maren sat in a chair that cost someone at Asterlyne a lot of points in a procurement spreadsheet—cheap enough that nobody would notice the line item, expensive enough to signal that this process mattered. Professional. Controlled. The kind of seating that said: *We take you seriously, but not too seriously, because if you were serious people, you wouldn't be here.*

The waiting room smelled like new plastic and the desperation everyone was pretending not to emit. Fluorescent ceiling panels buzzed at a frequency Cole could feel in his molars. A tablet on the low table in front of him displayed the charter—all 247 pages of it, not the 200 they'd claimed in the recruitment materials. He'd started counting on page one and hadn't stopped.

Around him, people were talking. Not talking, really. Performing the social shapes of talking—the woman to his left asking the woman next to her whether she'd done the medical screening, both of them knowing the answer was yes, both of them needing the distraction of words. The man in the corner—contractor's shoulders, calloused hands, not the type to volunteer for anything unless there was something urgent he was running from—staring at a tablet with the concentration of someone who'd decided not to read.

Cole read.

He had the luxury of being methodical. Nobody here knew him. Nobody was going to ask why he wasn't pacing the way the young guy near the door was pacing, or networking the way the two botanists in the corner seemed to be networking, trading names and institutional affiliations like they were at a conference instead of a one-way ticket off Earth. Cole could sit. Could read. Could let the fluorescent light strip the personality out of his face until he looked like he belonged here—which is to say, he looked like someone with nothing better to do.

The contract was written in English, which was kind. It could have been written in legalese—that subspecies of English that used all the same words but rearranged them to prevent understanding. But this contract was designed to be read. Just not *by* the signatories. The structure was forward-facing, user-friendly, the language was clear, the sections labeled sensibly. A person reading it in good faith would arrive at a different understanding than a person reading it carefully.

Cole read carefully.

Section 1: Organizational Charter of the Free Colonial Initiative, Parcel 16b Settlement.

All colonists agreed to the terms without qualification, without amendment, without negotiation. Fine. Nobody negotiated with Asterlyne. That wasn't the system.

Sections 2–5: the basic infrastructure. Settlement boundaries. Resource allocation. Command structure. A temporary governance council elected by majority vote, authority subordinate to the Company Safety Officer, who retained final decision-making on all matters of environmental management and resource distribution.

The woman next to Cole glanced down at his tablet. He felt her looking. He didn't move, didn't look back, just kept reading. She went back to not reading hers.

Section 6: Ownership and Land Rights.

This was where people usually started skimming.

Upon successful establishment of self-sustaining agricultural capacity and the passage of five (5) Earth years, title to Parcel 16b shall transfer to the colonist collective, contingent upon maintenance of charter compliance and population threshold stabilization.

Cole read that sentence three times.

Population threshold stabilization.

He flipped to the appendices. Nobody flipped to the appendices. The appendices were where you found the definitions that made you understand what you'd actually agreed to.

Clause 9.4.1: Definition of "colonist collective."

Ownership stake in Parcel 16b (upon transfer) shall be distributed equally among all original signatories to this charter present and alive on Year Five, Day One, beginning at 0000 hours UTC+7. Signatories who voluntarily depart the settlement, fail to renew charter consent, or are deceased shall forfeit their stake and have such stake distributed pro-rata among remaining signatories.

Okay. Straightforward. Forty-seven of them now. If someone left early, the pie got smaller, but their slice got bigger. People understood that math. It was a motivator—every person who stayed made everyone else's stake slightly larger. Asterlyne was good at architecture like that. Creating incentives that looked aligned but actually severed the group's interest from their survival.

Clause 9.4.2: Stabilization Threshold.

Transfer of title shall occur only if, at Year Five Day One, no fewer than fifty-one percent (51%) of original signatories remain alive and present at the settlement. Should this threshold not be met, title shall revert to Asterlyne Corporation in perpetuity, along with all improvements and resources thereupon. No compensation to colonists or next of kin shall be due.

Cole set the tablet down.

He looked at it on the table like it had grown teeth.

Forty-seven people. Fifty-one percent of forty-seven was 23.97. Functionally, twenty-four. You needed twenty-four people alive in five years, out of the forty-seven who started, to own anything.

Twenty-three could die. That was the margin.

He picked the tablet back up. Read it again. He hadn't miscalculated—his math was exact, and that was the problem. Nearly half the people in this room could die, and the contract would call it a success.

Cole looked up from the screen.

The contractor was still not reading his tablet. The woman next to him was reading the same page she'd been reading five minutes ago. The botanists were talking about soil chemistry, their voices settling into the kind of comfortable technical discussion that meant they'd probably worked together before, probably had a history that didn't start in this room. Near the window, a younger guy—couldn't be more than twenty-two, twenty-three—was rereading the same section over and over, his leg bouncing with the kind of focused anxiety that suggested he was trying very hard to understand something he was afraid to understand.

Cole counted the room. There were the two botanists. The contractor. The woman next to him. The bouncing kid. A doctor, fifties, calm, reading with the measured attention of someone used to processing medical literature. A small cluster of three in the corner who seemed to know each other—energy efficient, probably academic, the type who came to this because they'd burned through their funding and a charter settlement sounded like a field opportunity. Four people he hadn't classified yet. The facilitator, who was polished and bored in exactly the way that suggested she'd watched forty-seven batches of desperate people sign away their rights to land they'd never own.

Forty-seven faces.

Cole looked at each one and asked himself the same question: How many of these people will be dead five years from now?

Not will choose to leave. Not will want to go home. *Will be dead.*

The contractor had an old scar on his jaw that suggested he was good at surviving. That was one point in his favor. The bouncing kid looked like he couldn't survive a bad meal, but youth was its own strategy—faster healing, better reflexes, less likelihood of pre-existing conditions. The doctor had calloused fingertips that came from years of something—surgery, probably. Surgical hands were attached to people who made complicated decisions under pressure. That usually meant longevity.

He was still counting when the facilitator stood.

"We're ready to begin," she said. Her name was Melissa. It was on her badge. She said it the way you'd say: *We're ready to start the experiment.*

They were ushered down a hallway into a smaller room with individual signing stations—corporate ergonomics, each one separated by a divider so nobody had to watch anyone else sign. Privacy theater. Nobody needed privacy to sign away their rights, but it was more comfortable for everyone if they pretended they did.

Cole took his station. His tablet displayed the full charter, pre-populated with his name: COLE MAREN. His background check was complete. His medical clearance was already noted. His initial investment of fifty thousand credits—everything he'd been saving, everything he still owed on the loan that made the fifty thousand possible—was documented. His signature would make it binding.

He didn't sign immediately.

He ran through the math again.

Twenty-four alive. Fifty-one percent. Year five. Five years from day one until the clause triggered. Not five years from now—five years from the day they landed, from the moment the shuttle touched down on Parcel 16b and they became subject to the full weight of the charter.

In five years, Cole would be forty-three. Lily was nine. He'd had custody three years ago, back when the logistics job was still a logistics job and not a series of contract positions that paid less each quarter. He'd held custody like a trophy he'd won and didn't know how to carry, had let it slip through his hands the way he let everything slip eventually—good at systems, terrible at presence.

The career contracted. The apartment shrank. A two-bedroom became a studio became a room he rented by the week. The visits went from weekly to biweekly to monthly to quarterly, each reduction feeling less like a loss and more like a correction.

His ex-wife had filed for full custody eighteen months ago. She hadn't needed to file hard. He'd already surrendered the space where being a father lived.

He thought about Lily at nine, probably taller now than he remembered. The last visit had been awkward. She'd drawn a horse in colored pencil and asked if he wanted to keep it. He'd said yes. Meant it. Left it on the table when he walked out because the recruiter from Asterlyne was waiting and he needed to close the door on something, anything, to stop the feeling that everything he touched fell apart.

The horse was probably in a recycle bin somewhere.

He'd gotten the recruitment email the next day. It had landed in his inbox like an answer to a prayer he hadn't quite had the courage to articulate.

Cole signed.

He used the stylus on the tablet and dragged his name across the signature box. Official. Binding. Witnessed by timestamp, biometric scan, satellite uplink. Legally immutable.

He'd just bet the forty-three-year-old version of himself that he would be alive on Parcel 16b, one of twenty-four survivors.

He'd bet everything he hadn't already lost—on clause 9.4.2.

On the fact that he could count.

On the fact that if you counted carefully enough, you could make the math work. You could read what was written even when nobody wanted you to read it. You could understand what was being asked of you even when nobody would articulate the question.

Twenty-three people. That's what it cost.

He wasn't hoping nobody died. That would be stupid. People always died. Disease, accident, bad luck, the simple arithmetic of time. On Earth, people died too. At least here, the math was clear.

He walked out of the signing room into the hallway where the other forty-six people were filing out of their own stations, each one carrying the same tablet, each one having just signed the same charter, each one presumably having read the same clause.

Or not. Probably not.

Most of them looked like they'd been reading the user-friendly sections, the parts that promised land, ownership, a fresh start. Most of them weren't looking at the man in the corridor the way Cole was looking at them—the way you might look at a group of assets and try to calculate which ones would hold their value.

The contractor from the waiting room passed him. Scarred jaw, calloused hands. His name tag read POLK, G. He glanced at Cole the way you look at someone when you're deciding if they're competition or just another body in the headcount.

"Hell of a thing," Polk said. Not really to Cole. To the corridor, to the fluorescent light, to the general condition of signing away everything you had left.

He walked on before Cole could answer.

Cole watched him go. Polk hadn't read the contract. You could tell from the way he moved—angry but unfocused, the posture of a man who'd signed on instinct rather than analysis. He thought the danger was the frontier. Didn't know the danger was the fine print.

The corridor emptied. The fluorescent hum settled back into Cole's molars.

He'd signed up for the frontier. He'd actually signed up for the margin—the twenty-three lives between ownership and nothing.

And he was betting everything he had left—which wasn't much, which was the point—on the fact that he could count.

Chapter Two — Transit

Three months in recycled air tastes like someone else's exhalation mixed with industrial scrubbers. Cole learned to stop noticing it by week two, which meant he'd learned to stop noticing a lot of things. The Asterlyne transport vessel wasn't designed for comfort. It was designed for efficiency and just enough habitability to keep the cargo alive until delivery.

The crew of forty-seven had settled into the lower decks like sediment. Bunks in columns of four. A common mess that doubled as the orientation room. A medical bay where Haruki Sato—the doctor, the calm one, the man who'd introduced himself by asking everyone their medical history instead of their name—had already organized a basic clinic. Exercise equipment that nobody used because moving in 0.7G made the stomach betray you.

Cole spent most of his time reading.

The charter existed in full digital copy on the ship's network, along with supplementary materials: geological surveys, atmospheric data, agricultural projections, structural schematics. Personnel files. Cole started with the geological surveys and worked outward — atmospheric composition, seasonal modeling, soil chemistry baselines. He read the way he'd always read corporate data: not for what was there, but for what should have been. The surveys were clean. Too clean. Methodology notes referenced comparison datasets that weren't included. Soil analysis tables started mid-sequence, as though earlier measurements had been stripped. The gaps were small and the gaps were deliberate.

It was the kind of thing you'd notice if you knew what missing data looked like. Most people didn't.

Most people were doing what they were doing in the common mess on the forty-fifth day of transit: clustering into smaller groups, forming the soft hierarchies that would become the harder hierarchies once gravity meant something again. The two botanists from the initial signing — colleagues from an Earth research institute, both women, both operating under the assumption that the settlement would have built-in scientific infrastructure — had claimed a table and were planning projects. Three academics who didn't quite know each other yet but recognized the type — the kind of people who'd chased this opportunity because their funding had ended, because they had nothing else to lose, because an off-world colony looked like a second act.

Garrett Polk had gathered about eight people around him at the far table and was talking about freedom. Not freedom as an abstraction. Freedom as a wound.

"...systems that look rational," Polk was saying, his voice the kind that filled a room whether you wanted it to or not, "built by people who've never had to live inside them. You do everything right. Follow the plan. Then you find out the plan was never yours. It was for the people who wrote it."

Eight heads nodding. Cole watched from across the mess and heard the specific bitterness underneath the rhetoric — not ideology, but damage. The anger of people who'd played by rules that turned out to be traps, who'd built things that got taken, who'd signed deals that looked like opportunity and turned out to be architecture. Polk wasn't selling a philosophy. He was selling the feeling of finally being right about who had broken you.

Ren Tsai was at a table alone, reviewing something on her personal tablet that made her frown.

Cole had learned her name from the orientation materials. Engineer. Specialty: structural systems and environmental management. She'd been quiet during the embarkation process but had asked three questions during the first orientation presentation that had made the facilitators visibly uncomfortable: questions about the gap between the projected resource allocations and the documented planetary volatility, questions about failover systems if primary water processing failed, questions about what happened to a settlement if the Company Safety Officer position became unexpectedly vacant.

The facilitators had given the kind of answers that were technically true and completely unhelpful.

Ren had written down their answers anyway, which meant she understood that documentation mattered more than satisfaction.

Cole carried his meal tray to her table. It was a violation of the soft social rules—you didn't join someone sitting alone unless they invited you. But Cole was good at calculating when rules could be bent.

"Can I sit?" he asked.

Ren looked up. She was older than he'd initially thought—maybe mid-thirties instead of late twenties. Her face was the kind that stayed still when it was thinking, which made the moments when it moved mean something. It moved now.

"Sure," she said.

Cole sat. The chair was the kind engineered for low-G comfort, which meant it felt like sitting on a marshmallow someone had wrapped in steel.

"You asked good questions," Cole said.

"They didn't answer any of them," Ren said.

"That's usually how you know they're good questions."

Ren set down her tablet. Up close, Cole could see what she'd been reading: the resource allocation spreadsheet, compared to what looked like her own calculations. She'd been checking the math. Probably finding it didn't add up.

"What's your background?" she asked.

"Logistics. Inventory management, supply chain optimization, that kind of thing."

"So you count things."

"I keep things from running out."

Ren nodded like this explained something. It did. He could see her recalibrating what she was looking at—not just information anymore, but a problem. And problems needed people who could count.

"Why did you sign up?" she asked.

Cole had an answer prepared for this question. It was true and incomplete: "Earth didn't need me anymore. Out there, I might."

It was the kind of answer that closed conversations gently. Ren didn't push. But she looked at him the way someone looks at a calculation they don't quite believe yet, and he knew she was reading the deflection accurately. She understood that he'd answered a different question than the one she'd asked.

She didn't call him on it.

"I'm Ren," she said, and went back to her tablet.

Cole ate his meal — some kind of protein cultivate that had the texture of foam and the flavor of commitment to nutrition — and watched the common mess reorganize itself. The academics had drifted toward Polk's table, drawn by the gravitational pull of a man who sounded like he had answers. The botanists had separated, one heading to find Haruki in the medical bay, the other — Dana Wen — settling into a corner with her notebook. Actual paper. She filled pages in handwriting so small it looked like code. Who sat where. Who ate what. Which orientation modules people skipped. She avoided eye contact with the practiced ease of someone who'd learned that being overlooked was safety. Cole noticed her because she was designed not to be noticed.

Haruki was visible through the medical bay hatch, sitting across from a colonist in what had become his daily routine — colonists sat down one by one for conversations that started as medical histories and ended with questions nobody wanted answered. How long without your medication? What happens when resupply stops? He asked them in that measured voice, as though he were merely filling in forms.

Joss Carver — twenty-two, the youngest on the manifest — was panicking and preparing in roughly equal measure. He'd cycle through the charter with his leg bouncing, then pivot to the survival appendices, building checklists, cross-referencing agricultural projections with settlement schematics. He'd watched every orientation video Asterlyne had loaded onto the network. Asked Haruki about field wound care. Asked Ren about water system failures. Over-prepared in the way that young people sometimes are when preparation is the only thing they can control.

Polk was laughing about something at his table. Not a happy laugh. The kind that lived in the space between humor and anger, the sound of a man who'd learned to make jokes about things that had broken him.

By the sixty-third day, Cole had learned the rhythms of the ship well enough to move through it without thinking. He knew which paths connected the medical bay to the agricultural storage

without passing through the common mess. He knew when Ren went to the exercise equipment and came back all motion and sharp breathing. He knew that Dana did her heavy work in the agricultural lab in the early hours, when nobody else was awake.

He'd organized his own analysis by then. Built a quiet spreadsheet in a private partition of the ship's network. Listed forty-six names — Garrett Polk already filed from signing day, the man who'd said *Hell of a thing* to the corridor and kept walking, all that anger with nowhere to land. Coded them by skill set, by apparent psychological resilience, by the things they'd admitted in casual conversation about why they'd signed.

It was a monster's work, really. Ranking people by survival probability.

But Cole had stopped caring about being anything other than useful.

Polk found him on the observation deck three weeks before planet-fall. The deck was the one section of the ship that looked out at nothing — vast dark, occasional stars, Parcel 16b visible only as a brightening dot among fixed lights.

Polk was at the deck's single table, writing on a napkin. Not words. Numbers. Cole could see the shape of it from across the space — columns of figures, credit amounts, interest calculations. The arithmetic of debt. Polk was comparing what he owed on Earth against what the charter promised, drawing lines between numbers that didn't connect, running the math the way you'd run it if you were looking for an exit that didn't exist.

Cole watched. Didn't hide it.

When Polk realized he was being observed, he crumpled the napkin. The instinct was immediate — not guilt about the numbers, but guilt about being seen running them. His face rearranged itself into the big, public expression Cole had watched him wear in the mess. But the crumpled napkin was still in his fist.

"You read the whole thing, didn't you?" Polk said. "The charter."

"Yes."

"The appendices. The sub-clauses. All of it."

"All of it."

Polk leaned back. Studied Cole the way he studied people — quickly, with the intelligence he tried to keep underneath the volume. Cole could see the question forming. Not the one Polk was about to ask, but the one underneath it. The one that had been on the napkin.

"Find any way out of here?"

Cole considered the question. The honest answer was complicated. There was no way out — no early exit, no loophole that dissolved the contract, no clause that made the debt calculations on Polk's napkin come out different. But there was clause 9.4.2, which wasn't a way out. It was a way through.

Five years. Twenty-four alive. Ownership in perpetuity. Polk didn't know about it. Hadn't read far enough to find it.

"No," Cole said.

Polk's face changed. Not defeat — something closer to relief. He lit up the way people light up when someone confirms what they've been afraid to believe: that the game really is rigged, that the math really doesn't work, that it isn't just them.

"No," Polk repeated, almost to himself. "No, there isn't. There never is." He smoothed the crumpled napkin against his thigh, then pocketed it. "I knew it before I signed. Built a whole business on numbers once. Did everything right. Numbers worked. System didn't." He looked out at the dark. "You run numbers long enough, you learn they only work for the people who wrote the equation."

He stood. He was done. He'd gotten what he came for — confirmation from the one person on the ship who'd actually read the contract that the contract was exactly what he already believed it was: a trap you couldn't calculate your way out of.

Cole watched him leave. Understood three things: Polk hadn't read the charter. Polk thought Cole was a fellow prisoner in resignation. And Polk had no idea that resignation and strategy look identical from the outside but are entirely different machines.

Cole didn't correct him. Didn't mention clause 9.4.2. Didn't tell him there was something in the appendices worth finding.

Some information was only useful if the right person had it.

He turned back to his tablet. The weather projections. The agricultural failure vectors. The skills that would matter in months two and three and twelve when Earth was a memory and the supply drops became theoretical.

He wasn't resigned.

He was building the mathematics of survival.

The final orientation briefing happened five days before planet-fall. All forty-seven colonists gathered in the common mess. A facilitator—not Melissa from the signing station, someone new, someone even more polished and bored—presented the settlement overview.

Parcel 16b. Coordinates. Atmospheric composition. Seasonal cycles. Agricultural viability index. She said the word *index* with the precision of someone reading from a tablet, which is exactly what she was doing.

"The viability index for this parcel," the facilitator said, "has been measured at eighty-four."

Cole looked up.

Eighty-four.

He didn't need to check his tablet. He'd memorized the data appendix weeks ago — every table, every index, every footnote the supplementary materials had buried behind the recruitment brochure. The viability index in the actual survey data was seventy-two.

Seventy-two. Not eighty-four.

A difference of twelve points. The difference between optimism and actual capability. The difference between a projectable settlement timeline and a question mark.

He looked at Ren across the mess. She was looking at him, had been looking at him, had clearly noticed the discrepancy at the same moment and was watching to see if he'd registered it. When their eyes met, she looked away—not dismissive, just confirming. You saw it. I saw it. The question became: what were they going to do about a lie that was buried in supplementary materials that most people wouldn't read?

The facilitator continued her presentation, and Cole returned to his spreadsheet, to his calculations, to the mathematics of scarcity and survival.

The brochure had promised eighty-four.

The planet, he suspected, would deliver seventy-two.

And the colony would have to make the difference work with four thousand kilometers of distance and five years of solitude and the kind of mathematics that only looked simple until you had to live inside it.

Cole started a new column in his spreadsheet.

He labeled it: *Margin for Error*.

He left the cell empty.

He didn't know yet what number should go there, or even if any number was possible.

But he was going to find out.

Chapter Three — Parcel 16b

The descent came in three phases. First, the long brake through atmosphere, the shuttle shuddering and groaning the way metal groans when it's asked to defy physics and mostly wins. Second, the sight of it—planet swelling in the viewport from a dot to a thing, from a coordinate to a world. Third, the silence after the engines cut, when forty-seven people held their collective breath and tried to calibrate what they were actually seeing.

Cole watched the planet resolve from his seat, which had a window if you considered a reinforced polymer panel with limited viewing angles a window. Spiral-trunked trees dominated the landscape, their bark twisted like helical stairs climbing to canopy heights that made Earth timber seem practical. The light came through at an angle that suggested either early morning or some orbital geometry that wasn't standard, and the sky wasn't quite blue. Shifted toward green. Lavender-gray

soil showed through gaps in vegetation—not barren, just different, the color of something that had been dead once and stayed that way.

Beautiful. That was the thought most of the forty-seven were having, Cole could hear it in the soft sounds they made—not gasps exactly, more like the audible equivalent of holding still. Beauty was a reasonable response to seeing a world for the first time. Cole understood beauty. He could calculate the wavelengths. But he'd trained himself not to *feel* it as a category of data that mattered. Beauty made you make different decisions. Beauty made you take risks.

He'd already calculated enough risks.

"Descent protocol," the pilot announced, her voice crackling through the shuttle's speakers. "Ground contact in approximately four minutes. Secondary teams, confirm your equipment manifest."

Ren Tsai, who'd been sitting three rows behind Cole, was already moving—engineer's efficiency, not tourist curiosity. She was checking seals, verifying that the environmental equipment cases had survived transit intact. Haruki, the doctor, was walking the narrow aisle checking on colonists, making sure nobody was about to vomit or panic or do something that would make the landing harder. The contractor—Polk—was looking at the trees with an expression Cole couldn't quite read. Not afraid. Not happy. Something in between. Maybe it was relief. Maybe it was the look of a man realizing he'd traveled as far as you could travel and the things that broke him had all stayed on Earth.

The shuttle touched down at 0847 hours, local time. The notification came through Cole's tablet: *Parcel 16b Settlement Zone. Atmospheric pressure nominal. Radiation levels acceptable. Welcome.*

Welcome. As if the planet cared.

The descent ramp lowered, and the air that came up into the shuttle was wrong in three ways simultaneously. First, there was the ozone smell—sharp, electric, the kind of smell you got from high-altitude storms. Second, something sweet underneath it, something organic, rotting, alive all at once. Third, nothing. No sound. No birds, no insects, no ambient fauna noise. Just the hum of the shuttle's idle systems and forty-seven people breathing in unison.

Cole was the fifth one down the ramp.

The soil under his boots was springy—higher moisture content than the surveys suggested, or the vegetation had worked out a water table management system that kept things damp near the surface. The air was warmer than he'd expected, which was interesting because the thermal data had been pretty clear about day-night oscillation. This was either early in the warming cycle or the measurements had been taken at a different time of year and the planet's orbital dynamics created seasonal variance the viability index hadn't mentioned.

None of that was his immediate problem.

His immediate problem was the equipment manifest, which was being offloaded by a series of industrial loaders that had their own power sources and protocols and had all been checked by people in Asterlyne facilities who'd never been to Parcel 16b and would never face the consequences of their calculations being wrong. The manifest showed: four pre-fab shelters, rated for thirty occupants at minimum spacing standards. Forty-seven minus thirty equaled a problem that had been solved by

putting bunk beds closer together and hoping the structural engineers had built in safety margins they wouldn't tell you about.

Cole inventoried as the loaders set boxes down:

Medical supply cache—comprehensive list on the side panel, he skimmed it while pretending to do something else. Standard Earth medicines. Standard Earth diagnostic equipment. Quantities calculated for a baseline population of approximately twenty, contingency going to thirty. Forty-seven was not a number anyone at Asterlyne had planned for, or if they had, they'd decided the overage could be solved with mathematics that didn't include redundancy.

Agricultural supplies—seed stock in cryo-stasis, hydroponic nutrients, soil amendment compounds. Everything designed to jumpstart food production in alien soil that behaved according to specifications derived from a single survey. Dana Wen, the botanist, was checking the seed cases against her own manifests, her face doing that thing where it stayed still while something underneath it worked very fast.

Shelter composite and assembly supplies—enough to build the four structures that were currently being arranged in a grid by Polk and a cluster of other colonists who'd decided that starting work immediately was somehow less terrifying than standing still. The composite was a polymer-ceramic blend, rated for temperature oscillation within certain parameters. The parameters had a range. The night temperature data suggested they might breach that range.

Water extraction system—mechanical filters, UV treatment, mineral analysis equipment. The system was designed to process maybe 200 liters per day reliably. Cole had done the mathematics: forty-seven people, basic sanitation protocol, was approximately 100 liters daily. Comfortable margins. But that was assuming a consumption model where people showered on schedule instead of hoarding water, assuming the planet's hydrology was stable, assuming the filters worked the way they were supposed to work on water that had never been sampled in the field.

Assumptions were dangerous.

Ren materialized next to him while he was photographing the equipment layout with his tablet, building his own digital manifest against the official one. She was doing the same thing, Cole could see the photograph-and-verify rhythm.

"Medical allocation is short," she said quietly. Not a question, not a panic. Just a statement of mathematical fact. "About thirty percent of full redundancy."

"By design," Cole said. "Asterlyne was betting on low population stability. Figure if enough of us don't survive the first year, the remaining group becomes manageable within their supply calculations."

Ren was quiet for a moment. She was looking at the horizon, at the forest edge about a kilometer out, at the trees that hummed—not audibly yet, but Cole could feel something in the frequency range that made the back of his neck tense. A vibration. A suggestion of sound that wasn't quite sound.

"They knew," Ren said. "When they sent us out here with insufficient resources, they knew."

Cole had already factored this into his model. "The charter is structured to make us expendable. Until we hit the threshold. After that, we become valuable."

"So we survive not because they want us to, but because they suddenly lose something if we don't."

"Yes."

Ren looked at him. "That's a horrible system."

"It is," Cole agreed. "But it's a system. Systems are predictable. I'll take predictable horror over random chance."

The settlement assembly took most of the daylight hours. By 1600 hours, all four shelters were standing—not livable yet, still needing interior partitioning and equipment hookup, but standing. The colonists had fallen into work patterns naturally: the academics organizing a rough commons area, Polk and three others securing the perimeter with what the viability report called a "deterrent fence" and what Cole recognized as a gesture at safety that wouldn't stop anything actually determined to come through. Haruki was establishing the medical shelter setup. Dana was already into the soil, taking samples, running them through the portable analysis equipment.

Cole was finishing a grid map on his tablet, logging every resource location, every distance calculation, every vector for failure he could preempt with foresight.

The sun—wasn't quite a sun, not in the Earth sense, the light quality was wrong somehow, too green, too sharp at the edges—dropped toward the horizon sometime around 1830 hours. The temperature dropped with it, not gradually but in what felt like discrete steps. Cole checked his tablet. Readings dropping from 24 degrees Celsius at 1810 to 18 by 1830 to 11 by 1850. The rate of temperature loss was significantly steeper than the survey had projected. The survey had shown a gradual evening decline over approximately three hours. This was compression. Acute cooling.

Polk noticed it at the same time Cole did. He looked up from securing a section of perimeter fencing and registered the change, and Cole watched his face do the math in real time—contract promised certain conditions, reality was delivering different ones, those weren't bugs, they were features, and features couldn't be returned.

"Inside," Ren called out, already moving the stragglers toward the assembled shelters. "Everyone move inside the structures. Now."

The tone was different from the Earth tone, the one that had asked good questions during the briefing. This was the tone of someone who'd spent three months reading data and had just watched the data prove incomplete. This was the tone of someone whose job had just become impossible.

Forty-seven colonists filed into four shelters that had been rated for thirty.

The interior of Cole's assigned shelter—one of the four, himself in a bunk room with seven others—was a study in efficient space-wasting. The bunks were stacked in what was technically three-high but felt more like a very elaborate coffin installation. Minimal floor space. Communal sanitation in a corner that was going to create problems once more than two people decided to use it simultaneously.

The shelter's environmental system—basic climate control tied to the main power grid—hummed at a frequency that suggested it was working hard at a task it hadn't been designed for.

Night temperature dropped to 3 degrees Celsius by 2200 hours.

Cole lay on his bunk, in a sleeping configuration that was designed for people whose bodies were smaller than human bodies, and listened to the shelter groan. The structure was expanding and contracting as the temperature shifted. The composite was doing exactly what composite does when you ask it to hold rigidity through thermal stress: it was complaining. Not cracking. Not yet. But the sound was the sound of something being asked to do a job within specification while knowing it didn't actually have the margin.

He could hear the other six bunkmates breathing. Someone three bunks over was crying quietly—not a full breakdown, just the kind of soft sounds someone makes when they're trying to process that they've made a mistake and can't unmake it. Someone else was reading something on a tablet, the light bleeding through the darkness—probably the charter, probably looking for a clause they'd missed, a loophole that would suddenly make this sensible.

Cole didn't sleep. He lay in the dark and listened to the shelter and listened to the planet.

Around 0200 hours, the hum changed.

Not louder. Deeper. The vibration that had been in his neck all day was now in his chest, a kind of resonance that felt like it was coming from the ground. Cole pulled up his tablet and checked the seismic sensors he'd configured in the equipment deployment logs. Nothing. No seismic activity. This wasn't the ground moving.

This was something else.

He got out of his bunk—which took an awkward maneuver, bunks were not designed for dignified exits—and moved to the shelter's single observation port. The view was limited to maybe fifty degrees, but it was enough to see the forest edge.

The trees. The spiral-trunked things that had been beautiful in the descent viewport were different at night. They were still visible—there was a faint bioluminescence, something in the bark that glowed a very pale green—and the hum was coming from them. Not an orchestrated hum, not a unified sound. Individual trees, vibrating at slightly different frequencies, creating an interference pattern that wasn't sound so much as the suggestion of sound, the ghost of music that you felt rather than heard.

And something was moving in the undergrowth.

Not approaching. Cole watched carefully. Whatever it was, it wasn't advancing on the settlement. It was there, in the trees, in the shadows that the bioluminescent glow carved out, and it was *looking*.

Cole had spent three months building a spreadsheet of survival probabilities. Had calculated medical failures and resource depletion and interpersonal conflict cascades. Had modeled every variable the transit data offered.

The viability report said nothing about fauna. Not a word. The planet was supposed to be a soil chemistry problem, a weather problem, a supply chain problem. Nobody had mentioned the possibility that something would be out there in the dark, looking back.

He'd calculated everything except what it felt like to be seen by something that had evolved on a world that had never demanded humans exist on it.

The creature—and he was certain it was a creature, singular, patient, interested—watched from the forest. Cole watched back. Neither of them moved. The hours passed like that, both of them aware of each other's presence through the dark, through the humming trees, through a silence that had nothing to do with either of them.

By 0600 hours, it was gone. The forest was still there, still humming, still glowing faintly in the pre-dawn dark. But the sense of being watched released like a held breath.

Cole went back to his bunk. Lay down. The shelter groaned as the temperature began its slow climb back toward morning, and around him the other colonists stirred in the half-dark, shifting against bunks that hadn't been designed for sleep so much as containment.

He could hear them through the thin walls. Through the partitions. Forty-seven people breathing in a settlement that had been on this planet less than a day and was already learning what the planet thought of it.

Forty-seven.

He needed twenty-four.

The planet had been here for millions of years, cycling through seasons that nothing from Earth was designed for. They'd been here less than a day. Long enough to learn that the air was wrong, the data was incomplete, the shelters were insufficient, and something in the forest was watching.

Cole closed his eyes and did the only thing he knew how to do.

He counted.

Chapter Four — The Rules

By day four, the colony had stopped thinking of the planet as an antagonist and started thinking of it as a puzzle. By day seven, the distinction didn't matter anymore. Both puzzles and antagonists could kill you if you didn't pay attention.

Cole spent the first week building systems. It was what he knew — logistics, flow, allocation, redundancy. Warehouse management on an alien world. The same principles applied: figure out what you have, figure out what you'd burn through, calculate how long before the gap between supply and need started killing people.

Water collection came first. The diurnal temperature shifts were steeper than the survey data had suggested — Ren had measured a twenty-degree Celsius swing in twelve hours, double what the viability index predicted. Cole built a schedule around it: collection during the stable morning

window, containers sealed by mid-afternoon when atmospheric particles became more active in the cooling hours. Two people per station. Four-hour rotations. Clear protocols for filter type, storage method, contamination prevention.

Food rationing came second. The seed stock had gone into the lavender soil on day two, and by day five, the results were wrong in a way that didn't have an obvious explanation. Some rows took root and sprouted. Others, planted in identical soil ten meters apart, didn't germinate at all. A third set germinated, grew for two days, then wilted overnight like someone had switched them off. Dana Wen was spending twelve hours a day in the agricultural section, running the same tests on the same soil and getting different results every time. She hadn't explained the inconsistency yet, but the intensity of her focus said she was closing in. Until she did, Cole set three rationing tiers: full allocation for physical labor crews, reduced for technical and administrative, baseline for everyone else. Nobody liked the tiers. That was fine. Liking wasn't the point.

Shelter maintenance rotation came third. The pre-fab composite was warping under the UV cycle — Ren showed him the measurements, the thermal stress patterns that exceeded the rated delta every night. Cosmetic damage so far, but structural failure was trending toward month two. She'd drawn up a reinforcement plan that required redirecting labor from agricultural setup, which meant food production would slide, which meant rationing would tighten ahead of schedule. Everything connected. Pull one thread and three others moved.

Perimeter protocols came last, because Cole hadn't figured out what they were protecting against yet. The fauna in the tree line hadn't approached since the first night. The spore activity Haruki had flagged was nighttime-specific. The instability in the soil chemistry was slow, not acute. So Cole set basic rules: nobody beyond the markers after dark, buddy system for scouting parties, mandatory check-ins every four hours.

Systems. Structure. The mathematics of not dying.

The colonists hated it.

Not all of them. Some understood — the ones with technical backgrounds, the ones who'd read enough of the charter to grasp that Parcel 16b wasn't a homestead, it was a five-year survival problem with a contractual threshold. Ren didn't love the hierarchy but she respected the engineering logic. Haruki followed the protocols without comment, which was its own endorsement.

But at least a dozen of them hadn't come here for a shift schedule. They'd come for freedom — for the promise buried in Asterlyne's recruitment materials, the one that whispered: *Leave the systems behind. Start over. Be something other than what the old world made you.* And here was Cole Maren, two weeks on an alien world, already building the same architecture they'd crossed the void to escape.

Garrett Polk led the resistance. Not formally — Polk didn't organize. He just refused, and other people watched him refuse, and refusing became something you could do.

Cole found him at the eastern water station on day five, filling containers without the pre-filter. When Cole pointed it out, Polk didn't argue immediately. He just looked at Cole with something that went past disagreement into a place Cole recognized.

"The filters are slowing us down," Polk said.

"The filters are preventing us from drinking contaminated water."

"We don't *know* that." Polk set a full container down with the controlled force of a man managing his own hands. "We just think it is. Based on the survey. Same survey that said the shelters would handle the cold. Same survey that said the UV wasn't a problem. Everything they promised us was wrong, but we're still following *their* rules?"

"These are my rules," Cole said.

"That's supposed to make me feel better?"

Cole understood what he was hearing. Polk had built something on Earth — a business, a system, something that should have worked. The numbers had been right and the outcome had been wrong, and Polk had lost everything while following rules someone else wrote. Now Cole was standing in front of him with another set of rules, another set of numbers, another architecture that looked rational and promised survival. Polk wasn't seeing a water protocol. He was seeing a trap dressed up as salvation. The same trap, different planet.

When Cole explained the math behind rationing — the caloric intake calculations, the projected agricultural timeline, the consumption models that kept everyone breathing to the five-year mark — Polk listened. Heard every number. And the logic reinforced everything he already believed: that math was the language people in charge used against people like him. Not because the math was wrong. Because the math was right, and it didn't matter.

"I'm not staying here long enough for contaminated water to matter," Polk said. His voice had gone flat — not angry anymore, just settled into the conviction that the game was rigged either way. "Hoping Earth sends rescue by Year Two. Until then, I'm making this livable, not perfect."

Cole documented the deviation and moved on. But he'd seen it now. The math didn't persuade Polk. The math was the problem.

Cole was still modeling contingencies when Suri Park ate the fruit.

She was one of the ones he'd filed as useful-dangerous. Late twenties, compact, curious. The kind of person who asked questions rapidly and then acted on the answers before fully thinking them through. Useful in exploration. Dangerous in an environment where exploration was still a luxury.

Day thirteen. She'd walked the southern perimeter, mapping the immediate flora, checking what looked edible against the preliminary agricultural database. Found berries clustered on something between a shrub and a low tree. The database showed them safe. Nutrient profile good. No known toxins.

The database had been built from a single survey conducted during a single visit.

Suri ate one.

The reaction came forty minutes later. She was back at the settlement, documenting her finds, when her hands started to shake. Not from cold, not from fear — from something her nervous system was

registering as a profound threat. Pupils dilated. Breathing accelerated. Within an hour, she was seizing — full tonic-clonic, her body rejecting a compound that the database said didn't exist, her immune system fighting something Earth medicine had never catalogued.

Haruki got her to the medical shelter. On her side to prevent aspiration. Muscle relaxant from the pharmaceutical stock — shouldn't have worked on an alien compound, but it slowed the seizure enough to keep her from breaking bones. He watched her through the worst of it with the steady hands of a man who had done this before and would do it again.

By day fourteen, Suri had stopped seizing.

Cole sat with Haruki while they waited for her to wake.

"She followed protocol," Cole said.

"She did." Haruki had his hands folded, still. "The database said those berries were safe. It said that based on readings taken during the survey window. Today is not the survey window."

"So the chemistry changed."

"The chemistry is always changing. On Earth we call it seasonal variation — compound expression shifts with temperature, light exposure, soil chemistry. A plant that's safe in June can be toxic in November." Haruki looked at Suri's sleeping form. "The difference is that on Earth, we've had ten thousand years to learn the calendar. Here, we don't even know the length of a season."

Cole sat with that.

The viability data wasn't wrong. It was incomplete. A single survey had captured a single moment, and they'd been navigating by it as if the planet would hold still. But the planet wasn't holding still. The soil chemistry Dana was tracking, the temperature swings Ren was measuring, the compound variations that had nearly killed Suri — all of it was moving, cycling, shifting on a schedule nobody had mapped.

They'd been reading a still photograph and calling it a map.

He went to find Dana. She was in the agricultural section, three soil samples lined up with a fourth added since morning. Same soil. Different readings.

"It's not random," she said without looking up. "There's a pattern. I don't know what's driving it yet, but the shifts aren't noise. They're signal."

"The berries Suri ate."

"Compound concentrations have shifted since the survey. Enough to cross from safe to toxic."

"How long until you can predict the shifts?"

She looked at him. Her expression was the careful blankness of someone running calculations she wasn't ready to share. "I need more data points. Weeks, maybe. If the pattern holds."

"And if it doesn't."

Dana went back to her samples.

Cole called the settlement together that evening. Not a formal assembly — people gathered by the common area between shelters where they'd been eating meals. He told them about Suri. Told them the database was unreliable. Told them the new protocol: nothing native consumed without explicit clearance from both Dana and Haruki. Everything filtered. Everything tested. Everything assumed dangerous until proven otherwise.

Polk spoke first. "So the rules change again."

"The planet changed the rules," Cole said. "I'm keeping up."

"You're good at rules, Maren. Bet you were good at them back home too."

Cole let it land. Polk wasn't entirely wrong. Cole did think in systems. He did impose structure. He did believe the mathematics of organization was the margin between survival and entropy. What Polk missed was the assumption that Cole enjoyed it.

Ren stood. Not confrontational. Just clear.

"The survey was supposed to answer these questions," she said.

"It did," Cole said. "For a specific moment in time."

"Then we need to build our own answers."

That was the cleanest thing anyone had said since landing. Cole let it settle over the group — the idea that they weren't inheritors of someone else's data anymore. They were the survey now. Whatever they learned, they'd learn by living through it.

The gathering broke apart. People went back to shelters, to tasks, to the quiet work of processing that the planet they'd signed up for was not the planet they were on.

Cole stood at the perimeter as the temperature began its drop. The forest hummed. The soil under his boots was springy, responsive — alive in ways he couldn't see and was only beginning to measure.

The planet had rules.

They just hadn't finished learning them yet.